
Why Eating Well Feels Like Nothing

Designing De-Moralised Feedback Infrastructure for the Transition from Medicine 2.0 to Medicine 3.0

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IRP Report — Proposal (Part 1) + Realisation (Part 2) · June 2026

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Statement of Authorship and Acknowledgements

This report was written independently as part of the Independent Research Project (Design Futures, Royal College of Art). The research direction emerged from a sustained personal inquiry into preventive health, dietary behaviour, and the design of feedback systems, grounded in an extended auto-ethnography of my own eating and metabolic self-observation.

I am grateful to my IRP tutor for critique across Term 2 and Term 3, which sharpened two things in particular: the ethical treatment of eating-disorder risk and Food Freedom as hard constraints on any feedback design, and the discipline of narrowing a deliberately broad systemic inquiry into an actionable, audience-specific project. I also thank peers from the April 2026 critique exhibition for feedback that strengthened the macro-societal framing of the work.

Use of AI Disclosure

This submission was produced by me, in my own words and using my own imagery, except for quotations and images from published and unpublished sources, which are clearly indicated and acknowledged. The source of any picture, map, diagram or illustration is indicated, as is the source of any material not resulting from my own observation.

I used AI in the following ways:	(i) systematic identification of behavioural-science and metabolic-health literature; (ii) structuring, copy-editing and alignment of my own draft text against the report template; (iii) rendering of data visualisations and diagrams from my own data and concepts. All
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critical analysis, conceptual framing, the double-helix model, scenarios and speculative directions are my original work.

I used the following prompts:

(To be completed: list the 2-3 key prompts used.)

Print name: Weijie Li Signed: _____ Date: _____

Part 1: IRP Proposal

Abstract

Dietary behaviour change is hard to sustain because the benefits of eating well are largely imperceptible on everyday timescales: a week of good eating produces no intuitive score and often no felt sense of progress. This report reframes the resulting intention–action gap not as a motivation deficit but as a **broken feedback infrastructure** built on two coupled opacities — *food opacity* (we cannot see what processed food contains or does) and *body opacity* (we cannot sense our own metabolic response; one can be metabolically ill without being visibly overweight). Prevailing self-tracking paradigms (manual logging, evaluative dashboards) fail to produce durable engagement and can amplify harm pathways linked to disordered eating, because they layer **moral judgement** on top of data.

The project’s central design move is therefore *de-moralisation*: making the invisible visible through feedback that is descriptive, baseline-relative and navigational — like a sat-nav that reroutes without scolding — rather than rejecting data altogether. Situated at the intersection of behavioural design, speculative/critical design and futures studies, the work models the preferred future as a **double-helix feedback infrastructure** in which a user-side *Demand Arm* and an industry/system-side *Supply Arm* co-evolve, carrying society from reactive **Medicine 2.0** to proactive, personalised **Medicine 3.0**. It develops the REFLECTIVE heuristic and a 2050 speculative interface (Healthspan Curve / Food Architecture Radar / Neutral Readable Layer), and focuses on a defensible audience: a vast, largely unaware metabolically-vulnerable public, with the diagnosed pre-diabetic cohort as its sharpest-need apex.

Keywords: dietary feedback; feedback infrastructure; de-moralisation; Medicine 3.0; pre-diabetes; speculative design; interoception; ethics

1. Introduction and Justification of the Topic

I am interested in the future of preventive health behaviour — specifically, the future of **dietary feedback** for individuals trying to eat better in an increasingly opaque food environment. Unlike exercise or sleep, the benefits of good nutrition

are largely invisible on everyday timescales: a week of dietary adherence returns no intuitive score and little perceivable progress. This is not fundamentally a motivation deficit; it is a **structural feedback problem**. I call it the *phenomenological silence of eating well*.

This silence sustains a persistent intention–action gap. Present bias systematically disadvantages behaviours whose payoffs are delayed (Kahneman, 2011), while dominant interventions — manual logging in MyFitnessPal/Noom-like systems — impose high cognitive friction and return emotionally detached numbers rather than meaningful reinforcement. Health and fitness apps retain only 3–4% of users by Day 30 (Business of Apps, 2024), indicating that the prevailing interaction paradigm is not robust.

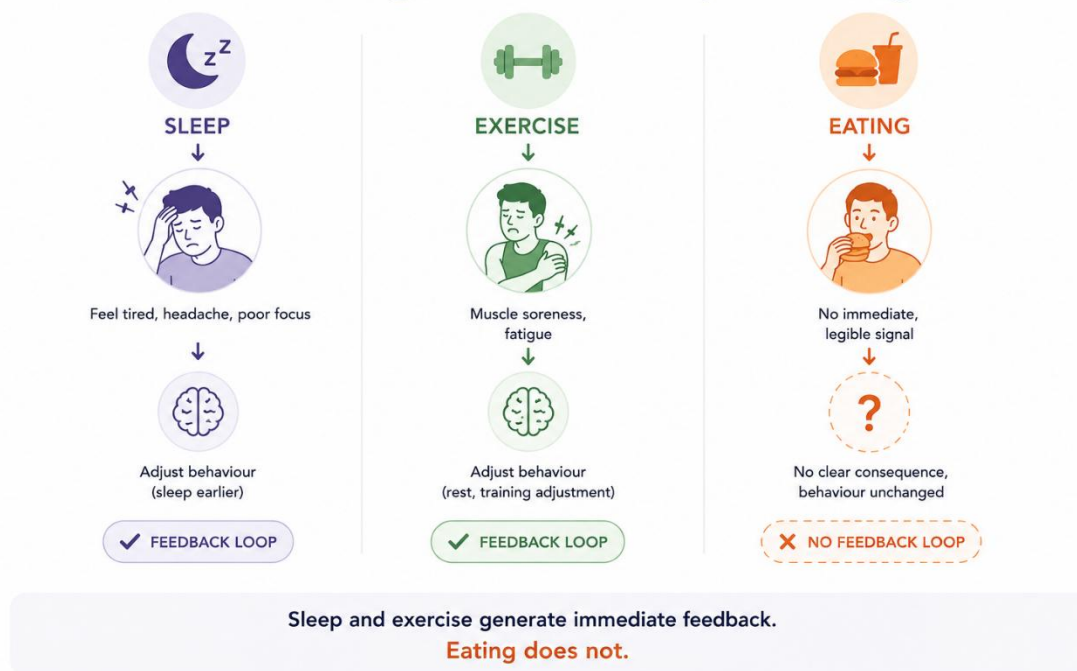
The stakes extend beyond engagement. Evidence links punitive self-monitoring to harm pathways associated with disordered eating: calorie-tracking tools negatively influence users with eating disorders (Phelan et al., 2020), and intensive self-monitoring is associated with elevated eating-disorder symptoms in non-clinical populations (Roth et al., 2024). NHS hospital admissions for eating disorders rose 8.5-fold between 2013/14 and 2020/21 (NHS England, 2023). Wallace et al. (2025) describe a plausible mechanism — the *reification of data*, the progressive replacement of direct bodily experience with quantified metrics. These findings make harm-prevention a **primary** design requirement, not a downstream mitigation.

Crucially, the problem is not data itself but the **moral grammar** wrapped around it: the silent slide from “your glucose rose” to “you were bad today”. The project’s stance is therefore *de-moralisation* — keeping data legible while stripping the judgement that turns measurement into self-punishment. This reframing lets the work embrace, rather than reject, the coming wave of consumer metabolic sensing.

A design-futures intervention is timely. The June 2024 FDA over-the-counter clearance of Abbott Lingo marks an inflection point: metabolic tracking is moving from clinical infrastructure into consumer products (Abbott, 2024). If non-punitive interaction principles are not articulated now, engineering-led, optimisation-first paradigms risk becoming the default vocabulary of dietary feedback. Positioned within a wider transition from reactive **Medicine 2.0** to proactive, personalised **Medicine 3.0** (Attia, 2023), the project asks how eating well might be made as legible and emotionally resonant as completing a run — under a strict ethical constraint: any feedback system must preserve food joy, avoid triggering disordered eating, and actively support **Food Freedom**, the right to make informed eating

choices without guilt or compulsion (Tribble and Resch, 2020).

The Missing Feedback Loop of Eating



2. Background: Literature, Practice, Signals and Trends

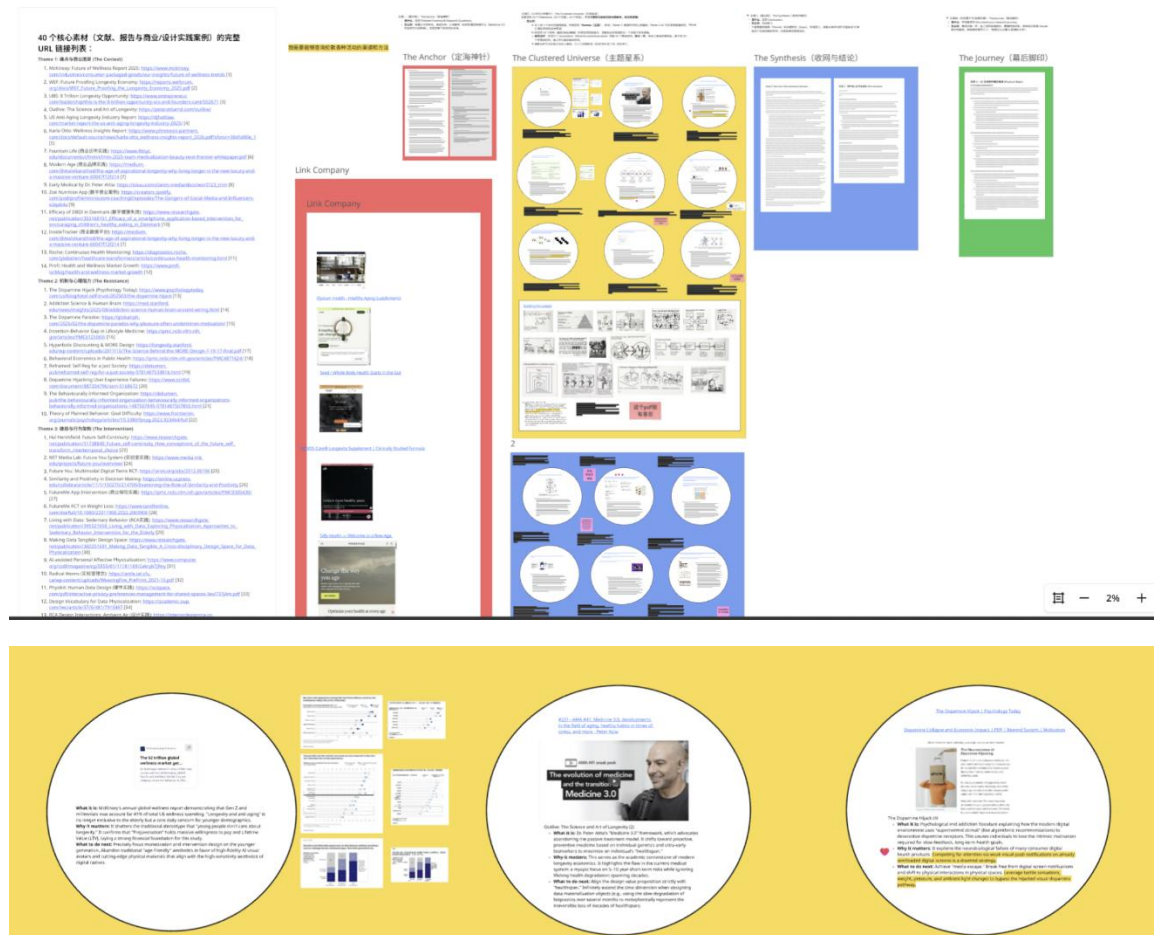
2.1 Research questions

1. Why does dietary behaviour produce so little immediate, legible feedback compared with exercise and sleep — for design, behavioural and physiological reasons?
2. Why do existing dietary-feedback tools fail to sustain engagement, and when do they actively cause harm?
3. What is the relationship between dietary-feedback design and disordered-eating risk, and can feedback be **de-moralised** to support Food Freedom rather than undermine it?
4. The feedback gap is not only individual: what **supply-side and systemic** conditions (food-industry incentives, opacity, policy) sustain it, and how might a demand-side and supply-side feedback loop co-evolve?

2.2 Research process

The review combined a visual literature-and-practice board (built across Term 2 and extended in Term 3) with an extended personal **auto-ethnography** and targeted **case-study analysis** of existing products and policy regimes. Sources span behavioural science, human-computer interaction, metabolic health, eating-

disorder research, food-systems policy and speculative design practice.



2.3 Research summary

The systemic failure of dietary tracking sits at the intersection of behavioural science and HCI. Fogg's Behaviour Model holds that behaviour occurs when Motivation, Ability and a Prompt converge (Fogg, 2009); in dietary apps, manual logging erodes Ability through friction while delayed summaries provide weak Prompts, so Motivation decays. Norman's *gulf of evaluation* explains why users cannot perceive the effects of eating well: metabolic benefits unfold over days to weeks, leaving little immediate evidence to evaluate (Norman, 2013). The phenomenological silence of digestion thus compounds present bias (Kahneman, 2011).

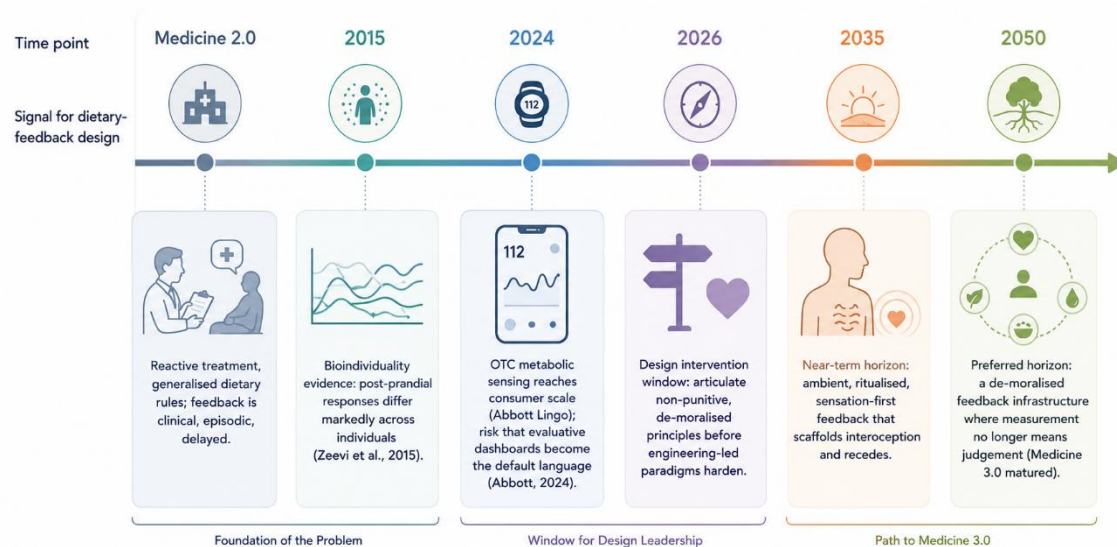
Attempts to bridge this gulf through intensive self-monitoring can induce harm. Wallace et al. (2025) describe the *reification of data* as a mechanism by which tracking interfaces displace bodily authority; meta-analytic evidence links intensive dietary self-monitoring to elevated eating-disorder symptoms (Roth et al., 2024). Intuitive Eating offers an ethical counter-model, cultivating interoceptive attunement and rejecting external rules in favour of internal hunger and satiety cues

(Tribole and Resch, 2020), with a protective association against binge eating (Linardon, 2020).

Physiologically, **bioindividuality** undermines population-average rules: identical foods elicit markedly different post-prandial glycaemic responses across individuals (Zeevi et al., 2015). This is the scientific basis for personal, baseline-relative, “after-meal” feedback — and for the project’s claim that there is *no absolute healthy food, only food suited to this body, now*. Two opacities make this illegible in daily life. **Food opacity**: ultra-processed foods are engineered around four industry goals — cheap, shelf-stable, palatable, scalable — that have no necessary relationship to metabolic health, and the resulting harms are diffuse and delayed (Lancet UPF Series, 2025). **Body opacity**: internal metabolic risk is hard to sense — the *TOFI* phenotype (“thin outside, fat inside”) shows that people of normal weight can carry dangerous visceral fat and be metabolically unhealthy, detectable only by imaging (Thomas and Bell, 2012). One can, in short, be sick without being fat.

At a systems level the project is positioned within the transition from **Medicine 2.0** (reactive, population-average treatment after disease appears) to **Medicine 3.0** (proactive, personalised prevention that tracks early markers such as apoB and insulin sensitivity decades ahead) (Attia, 2023). The same transition is propelled by the consumerisation of sensing (OTC metabolic devices) and by a cultural backlash against diet-culture moralisation.

Time point	Signal for dietary-feedback design
Medicine 2.0	Reactive treatment, generalised dietary rules; feedback is clinical, episodic, delayed.
2015	Bioindividuality evidence: post-prandial responses differ markedly across individuals (Zeevi et al., 2015).
2024	OTC metabolic sensing reaches consumer scale (Abbott Lingo); risk that evaluative dashboards become the default language (Abbott, 2024).
2026	Design intervention window : articulate non-punitive, de-moralised principles before engineering-led paradigms harden.
2035	Near-term horizon: ambient, ritualised, sensation-first feedback that scaffolds interoception and recedes.
2050	Preferred horizon: a de-moralised feedback infrastructure where measurement no longer means judgement (Medicine 3.0 matured).



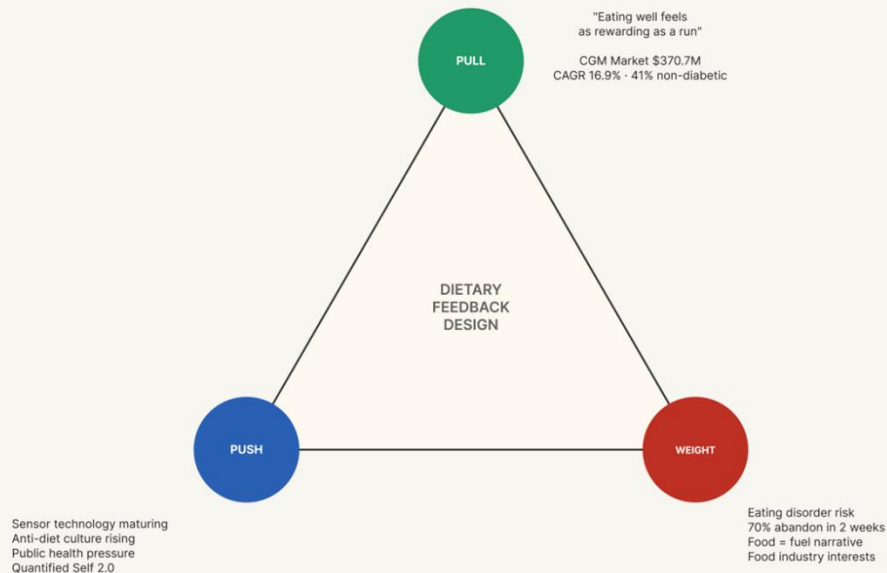
2.4 Conclusion — insights, trends, gaps

Three insights from literature and practice: (i) the dietary-feedback gap is a *design and physiological* problem, not a willpower problem; (ii) intensive tracking harms precisely when data is *moralised*, so de-moralisation is the pivotal lever; (iii) bioindividuality makes personal, baseline-relative feedback both scientifically valid and necessary.

Three trends/drivers: (i) consumerisation of metabolic sensing (OTC CGM); (ii) mainstreaming of Intuitive Eating / anti-diet interoceptive literacy; (iii) tightening food-systems policy (front-of-pack warning labels now adopted by 40+ countries; FDA's 2024 'healthy' rule) signalling a *supply-side* shift toward consequence disclosure (WHO, 2024; FDA, 2024).

Gaps: existing work over-optimises the *pre-meal* moment (rules, targets, good/bad labels) and under-designs *post-meal* sensemaking; almost all tools treat feedback as an individual problem and ignore the supply side; and few articulate how to make data legible *without* re-importing judgement.

What forces are shaping the future of dietary feedback design?



Framework: Inayatullah (2008) Futures Triangle · Adapted by Weijie Li, RCA GiD 2026

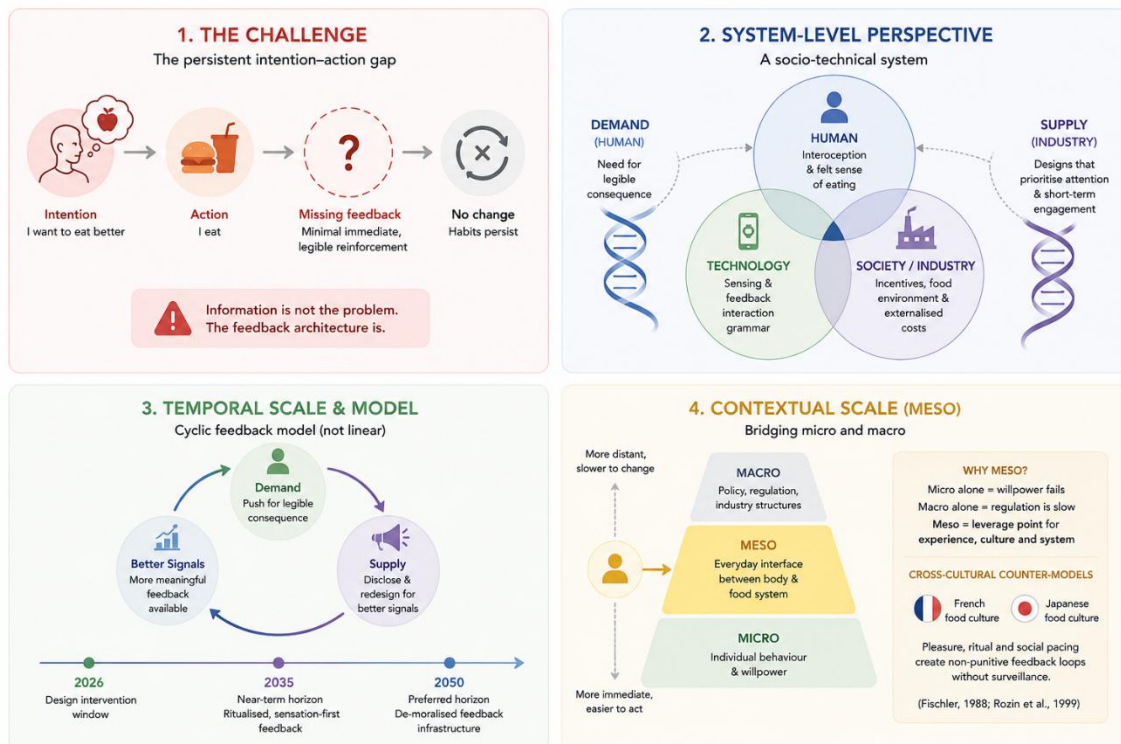
3. Problem Framing

The challenge. The persistent intention–action gap in dietary behaviour: genuine intentions to eat better routinely fail to become sustained change. The gap persists not because information is missing but because eating well generates minimal immediate, legible reinforcement — and because the food environment actively manufactures opacity. Dominant “solutions” add information and accountability without repairing the feedback architecture that makes nutrition feel experientially silent.

System-level perspective (human + technology + society). The project takes a socio-technical view. At the *human* level it concerns interoception and the felt sense of eating; at the *technology* level, the interaction grammar of sensing and feedback; at the *society/industry* level, the incentives that make “more tracking” the default product strategy while externalising long-term metabolic and relationship-with-food costs. The double-helix model (§14) makes the human (Demand) and industry/system (Supply) arms explicit.

Temporal scale and model. Medium-to-long horizon (2026 → 2035 → 2050), modelled **cyclically as a feedback loop** rather than linearly: demand for legible consequence drives supply-side disclosure, which improves the signals available to demand. This is why the central metaphor is infrastructure and helix, not product.

Contextual scale (meso, bridging micro and macro). The intervention sits at the *meso* level — the everyday interface between an individual body and the food system — deliberately chosen because purely *micro* (individual willpower) and purely *macro* (regulation alone) framings have both failed. Cross-cultural counter-models reinforce the meso framing: French and Japanese food cultures show that pleasure, ritual and social pacing can create non-punitive feedback loops without individualised surveillance (Fischler, 1988; Rozin et al., 1999).



4. Design Futures Approach

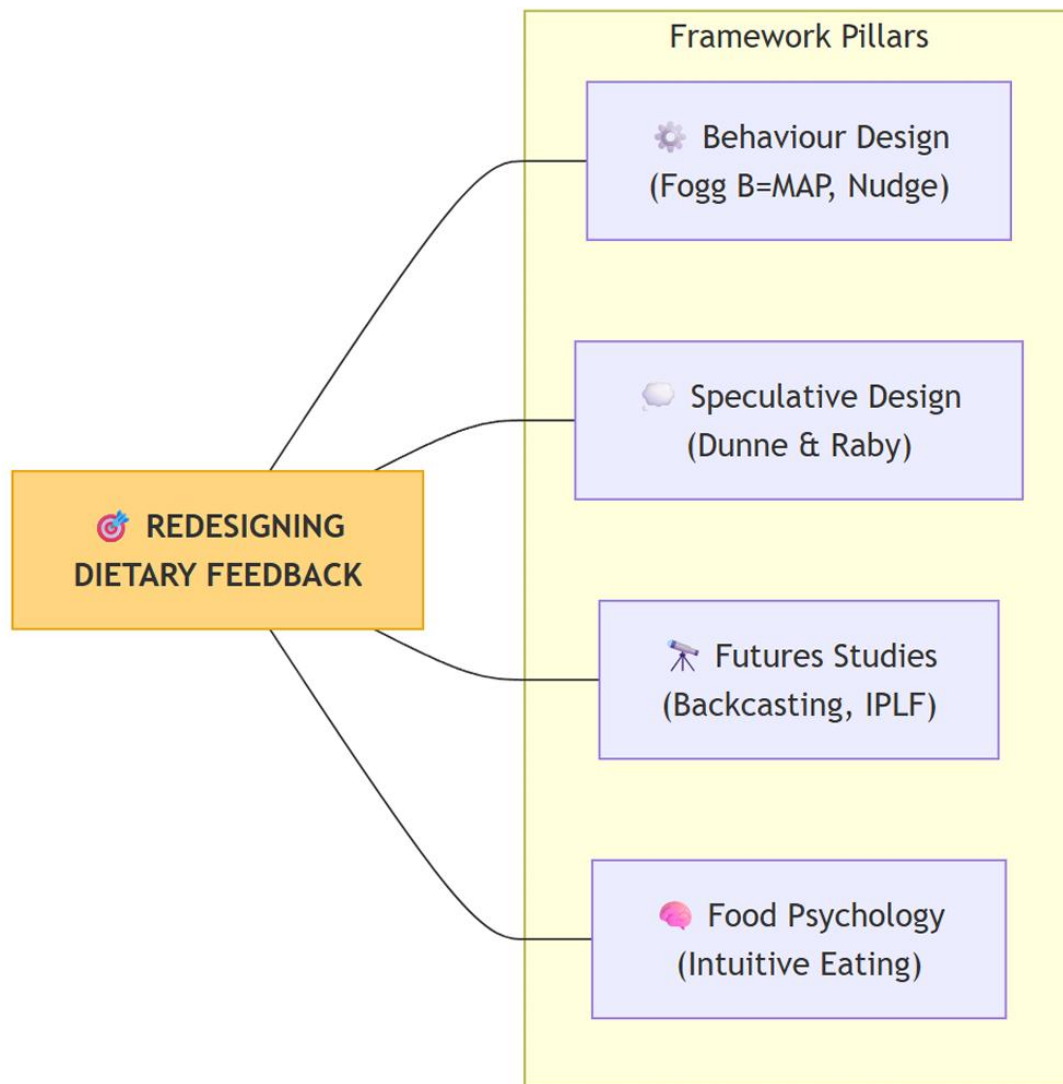
The project sits at the intersection of **behavioural design**, **speculative/critical design** and **futures studies**. My role as a design futurist is threefold: (1) *mapping the present* — assumptions, stakeholders and incentives that sustain the feedback gap; (2) *critical provocation* — artefacts that expose contradictions and absences (e.g. that “more data” can mean “more judgement”); and (3) *preference-led backcasting* from a 2050 horizon toward present intervention points.

Benefits, opportunities, limitations. Design’s benefit here is to make an abstract systems problem *tangible and arguable* — a ceramic disc or a speculative interface lets non-specialists handle and critique an ethical question. The opportunity is to set the interaction grammar of metabolic feedback *before* it hardens. The limitation is that speculative, small-sample work cannot establish clinical outcomes; it identifies safer principles and failure modes early, not efficacy.

Design approaches. The work draws on behavioural architecture (Fogg, 2009; Thaler and Sunstein, 2008), a compassion-based / harm-first orientation, and an

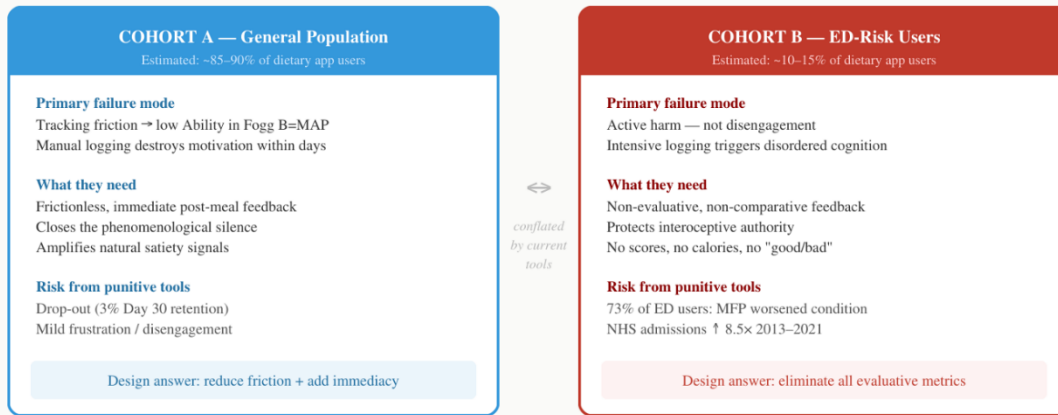
embodiment and empowerment stance that amplifies bodily authority rather than overriding it. These are operationalised in the **REFLECTIVE** heuristic: feedback should be *descriptive, not evaluative; relative, not absolute* (anchored to personal baselines; Zeevi et al., 2015); and *closed-loop, not open-ended* (a bounded post-meal closure rather than perpetual monitoring).

Futures approaches. The project uses speculative and critical design (Dunne and Raby, 2013) to “try on” alternative feedback logics, and strategic foresight tools — the Futures Triangle and Dator’s alternative futures (Inayatullah, 2008; Dator, 2009) — to structure scenarios as critique instruments rather than predictions.



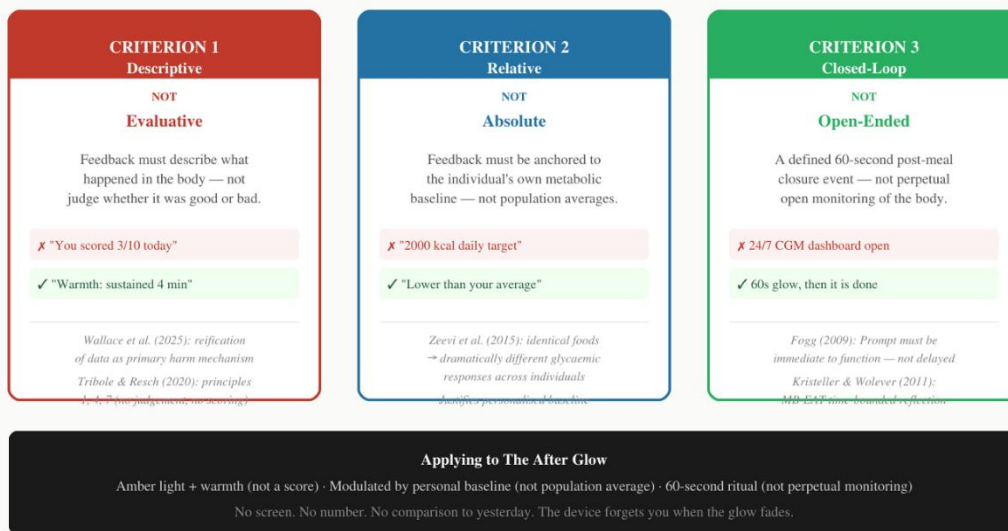
Two Distinct User Cohorts — Why One Design Cannot Serve Both

Current dietary tracking tools conflate these cohorts, causing harm to the second group



The REFLECTIVE Framework

Three evaluable design criteria for non-punitive dietary feedback systems



Framework developed in this project. Grounded in: Tribole & Resch (2020); Zeevi et al. (2015); Fogg (2009); Wallace et al. (2025).

4.1 Why this is a futures project, not a present-day one

A fair challenge — and one repeatedly raised in tutorials — is that dietary harm exists *today*, so why treat it as a *futures* project rather than a present-day intervention? The answer is that the problem and its solution-conditions are on different clocks. Following the test for a design-futures problem (an existing issue whose *trajectory* worsens, evidenced by trends and signals, with “roots going into the future”), the project maps the next three decades across four dimensions.

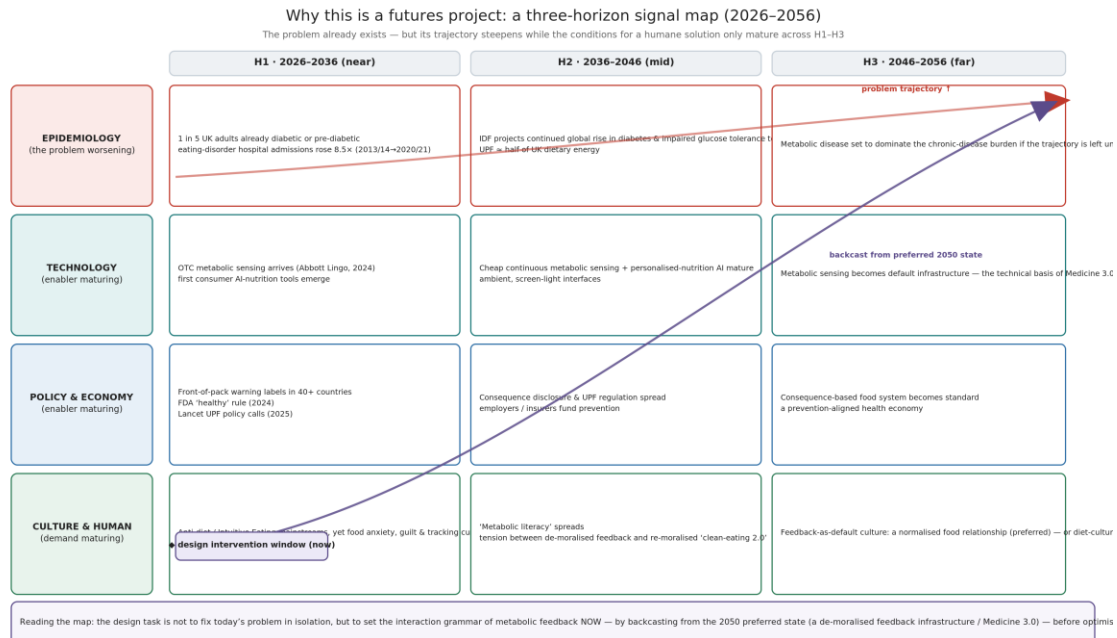


Figure 1 — A three-horizon signal map (2026–2056). The problem trajectory (top lane) steepens, while the conditions for a humane response — consumer metabolic sensing, consequence-disclosure policy, and a de-moralised food culture — only mature across Horizons 1–3. The design task is to set the interaction grammar now, by backcasting from the 2050 preferred state, before optimisation-first paradigms lock in.

Three points follow. First, the harm is **accelerating**: one in five UK adults is already diabetic or pre-diabetic, eating-disorder admissions rose 8.5-fold in seven years, and ultra-processed food now supplies roughly half of UK dietary energy — and the IDF projects continued global rise toward 2050 (NHS England, 2023; Diabetes UK, 2025; IDF, 2025; Lancet UPF Series, 2025). Second, the **enablers are immature**: consumer metabolic sensing only reached the market in 2024, consequence-disclosure policy is partial, and the culture is still moralising food — these mature over H1–H3, not now. Third, there is therefore a **closing window**: the interaction grammar of metabolic feedback is being written today, and if optimisation-first, diet-culture paradigms harden, the humane alternative becomes much harder to introduce later. The project's role is to backcast from a 2050 preferred state and intervene at that window — which is precisely what makes it a futures project.

5. Methodology

The methodology is **dual-track**: *investigative research* (to understand the gap and its failures) and *generative design research* (to prototype and evaluate de-moralised feedback), while holding a harm-prevention stance for ED-risk users throughout.

Data types. Primary, qualitative, experiential and visual data: my own extended **auto-ethnography** (somatic notes, post-meal observations, photographs); semi-structured interviews; photo-elicitation; and a mindful diary. Secondary data: literature, product **case studies**, and policy analysis. Qualitative, experience-near

data is appropriate because the research question concerns *felt sense* and *meaning*, not measurable efficacy.

Data sources and participants. Primary qualitative data from 6–8 adults aged 22–45 with prior dietary-tracking experience; current clinical ED diagnosis is an exclusion criterion to safeguard wellbeing; at least two participants from non-Western backgrounds to mitigate WEIRD sampling bias (Agapie, 2024). The auto-ethnography is a parallel, deep, single-subject primary source.

Collection protocol. Methods run across phases (Prime → Observe → Capture → Reflect → Attune → Build → Critique): phenomenological peer conversations; semi-structured and commensal contextual interviews; a 4-week mindful construal diary plus short post-meal EMA prompts; a sensitising probe kit; body-mapping warm-ups; speculative-artefact demos; and participatory critique using participants’ own photographs as stimuli. Documentation is kept through recordings, transcripts, a photo archive and a research log. Saturation is judged when new interviews stop producing new affect/agency themes.

Analysis protocol. Reflexive thematic analysis (Braun and Clarke, 2006), coding inductively for *affect*, *agency* and *tracking friction*, then relating themes back to B=MAP (Fogg, 2009) and Intuitive Eating as ethical constraint. To avoid “metric drift” (re-translating qualitative findings into new scores), synthesis privileges participant-authored, descriptive language (thermal, temporal, textural metaphors) as design material. Where primary data is thin, **case-study analysis** of existing tools and policy regimes is used to triangulate.

Design and futures methods. Scenario building, backcasting from 2050, and speculative prototyping (The After Glow; the 2050 three-layer interface) function as both outputs and research instruments.

Ethics and safety. Given evidence linking diet/weight tracking to ED risk (Roth et al., 2024; Phelan et al., 2020): clinical ED diagnosis excluded; questions framed phenomenologically; participants may skip or withdraw without penalty; no participant biometric data is required. Ethics checklist in Appendix C (assessed lower-risk).

5.1 Workshop and probe battery

Beyond the auto-ethnography and paired interviews, a battery of participatory probes is being run with the target audience (§13). Following tutor guidance to be candid about progress, the table below marks what has been conducted and what is planned for the remaining weeks — the gaps are part of the honest account, not hidden.

#	Probe / workshop	What it surfaces	Status
1	Baseline nutrition-literacy check	how much the public actually knows about basic nutrition	planned
2	Food-label	whether people can read pack labels and	planned

	comprehension task	additives — and the time, effort and understanding it takes	
3	Diet-history interview	which regimes people tried, whether they worked, what harm followed, and <i>why they trusted them</i> (influencer endorsement vs. evidence)	partly conducted
4	Body-literacy survey (+ booklet)	how well people understand their own body's signals	planned
5	Attitudes to data-monitoring	whether people welcome, fear or resist tracking, and whether they feel surveilled	planned
6	Ideal-future elicitation	a detailed description of the relationship with food people <i>want</i> — the food, the person, the purpose	partly conducted
7	Nutrition-education attitudes	whether people value nutrition knowledge and support teaching it in schools	planned
8	Food-choice priority ranking	how people rank value, price, taste and nutrition when choosing food	conducted
9	Data vs. senses preference	whether people would choose data, re-awakened bodily senses, or data-as-scaffold-toward-senses	conducted (informal)

The picnic commensal contextual-inquiry and the paired interviews (§11) have run; the survey-style probes (1, 2, 4, 5, 7) are scheduled. Probe 9 directly tests the project's central design question — *is data the goal, or a temporary scaffold toward bodily intuition?* — and early responses favour a “scaffold-then-recede” model over permanent tracking.

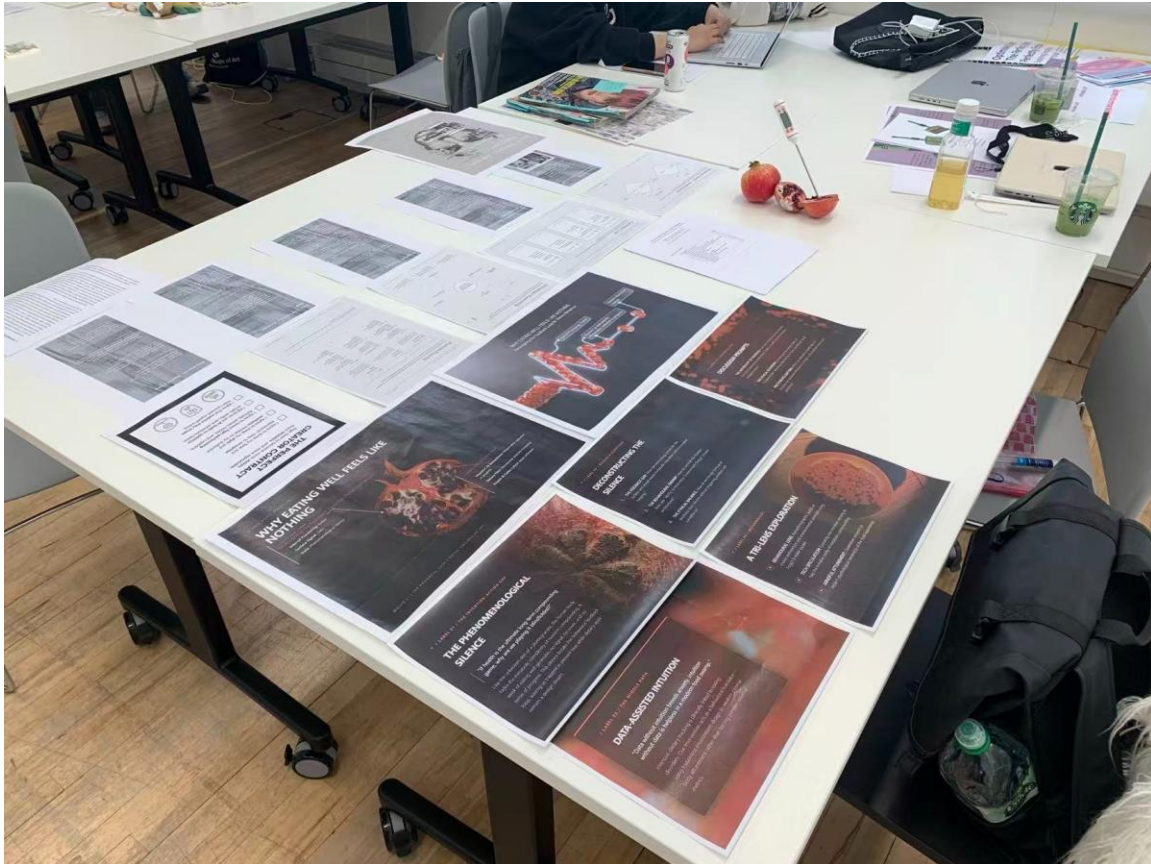
Participant welfare. Because these conversations can surface distress (several participants described anxious or restrictive eating), the protocol holds a duty-of-care stance: questions are experience-focused rather than evaluative, participation is voluntary and skippable, no participant is asked for biometric data, and participants are signposted to specialist support (in the UK, *Beat*, the eating-disorder charity) rather than treated as data points. This operationalises the harm-prevention commitment that the project treats as a first-order requirement.

6. Critique Documentation

The project was presented at the April 2026 group critique exhibition. Peer and tutor feedback highlighted three needs: (1) a stronger macro-societal frame linking the feedback gap to public health and market failure; (2) clearer separation of general users from ED-risk users; and (3) more tangible speculative outputs to ground theoretical claims.

The proposal responds in three ways. First, it strengthens the “why now” argument by linking the gap to a market inflection (OTC metabolic sensing) and to evidence of

harm when punitive tracking is normalised. Second, it formalises a two-cohort requirement — since refined into the **audience triangle** (§13) — treating ED-risk and high-need users as design stakeholders rather than edge cases. Third, it moves from argument to artefact: REFLECTIVE is paired with The After Glow and, in Term 3, with the systemic double-helix map and the 2050 interface.



7. Project Outputs, Stakeholders and Impacts

By the end of Term 3 the project will deliver: (i) the **Feedback Infrastructure Framework** (the double-helix model coupling Demand and Supply arms) as the project's conceptual contribution; (ii) the **REFLECTIVE** heuristic for auditing dietary-feedback tools against non-punitive, de-moralised criteria; (iii) a small **2050 scenario set** used as structured critique stimuli; and (iv) **speculative artefacts** — The After Glow and the three-layer interface (Healthspan Curve / Food Architecture Radar / Neutral Readable Layer).

Stakeholders now span both arms of the helix. *Demand side*: metabolically-vulnerable general public and the pre-diabetic apex (§13), health-tech UX designers, preventive-health clinicians and ED specialists (ethical gatekeepers). *Supply side*: food manufacturers, CGM/platform teams who shape default interaction grammar, regulators (front-of-pack disclosure), and employers/insurers incentivised by lower medical cost. The intended **impact** is a narrative and practice shift — from “more data is better” toward “feedback that is legible *and* de-moralised, protecting bodily

authority and Food Freedom” — and, systemically, toward consequence disclosure that lets demand and supply co-improve.

Stakeholder	Incentives / power	Design risk / design response
Users (general)	Want low-friction support; limited tolerance for manual logging.	Risk: abandonment from effort. Response: closed-loop, low-cognitive-load post-meal ritual (Fogg, 2009).
Users (ED-risk)	Need safety from moralised feedback.	Risk: data reification and compulsive tracking (Roth et al., 2024; Wallace et al., 2025). Response: descriptive, non-evaluative interactions; no scores/streaks.
Clinicians / dietitians	Duty of care; require interpretability and harm minimisation.	Risk: wellness claims without safety. Response: make REFLECTIVE auditable; include explicit safeguards (Section 5).
ED specialists	Ethical gatekeepers; can veto harmful paradigms.	Risk: “compassion-washed” surveillance. Response: anti-heroes method to surface failure modes (Appendix E).
CGM manufacturers	Control sensing infrastructure and default app patterns.	Risk: optimise for stickiness and scoring. Response: articulate alternative interaction vocabulary for CGM-era products.
Regulators	Constrain health claims and risk.	Risk: misleading claims/dark patterns. Response: keep feedback descriptive; avoid diagnostic language.
Venture / product leadership	ROI and retention incentives.	Risk: dependence by design. Response: design for off-boarding; “tool that disappears” end-state.
Food industry	Marketing and behavioural influence.	Risk: co-opting anti-diet language. Response: scenario critique makes these dynamics discussable without endorsement.

Table 3: Stakeholder map: incentives, risks, and corresponding design responses.

7.1 Output suite (educate · imagine · disseminate)

The outputs are developed across three registers, each mapped to the double helix (§14):

(1) Critical-design booklets — *educate* / *awareness*. A set of print artefacts that make food and body opacity tangible. The centrepiece is a **two-colour decoder booklet**: viewed through a red lens it shows the diet-culture *belief*; through a blue lens, the underlying *evidence* — dramatising the gap between what we think food does and what it actually does. Companion booklets debunk common diet myths (for example, “does eating after dark necessarily cause weight gain?”), pose “how

much do you really know about your body / your food?”, and include a **body energy-supply timeline** showing how the body is fuelled across the day. These target the demand-side literacy gap surfaced in §11 and the probe battery (§5.1).

(2) App demo — *imagine*. A working demonstrator of the de-moralised navigator — the three-layer interface in Fig 5 (resilience curve / food-fit radar / neutral reroute). It is explicitly a *near-future imagination*, not a shipping product, and openly lacks a real personalised database — a candid limitation that keeps it in the register of design fiction rather than a medical claim.

(3) Future-vision film — *disseminate*. A short video of the preferred 2050 future (Theo’s day, §15), to communicate the de-moralised feedback infrastructure to non-specialist audiences beyond RCA2026.

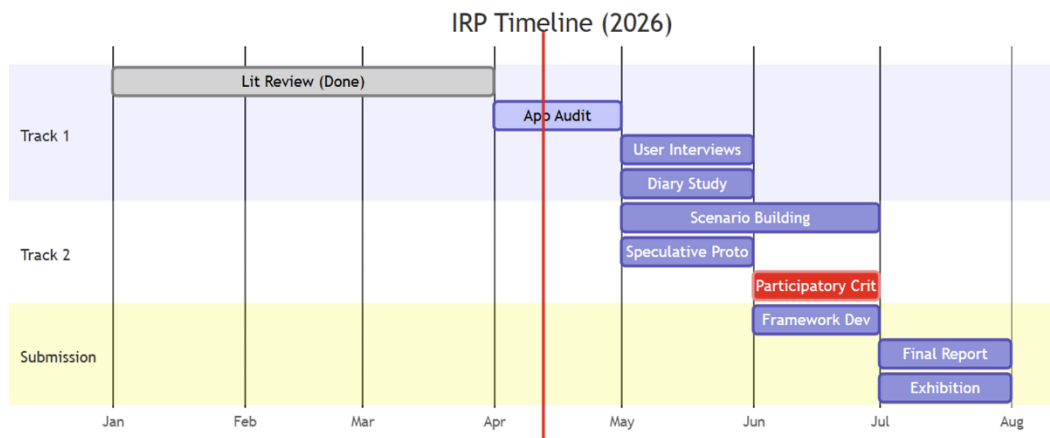
Together: the booklets and film grow demand-side literacy and cultural permission; the app models the demand-side interface; and all three argue, by example, for the supply-side shift toward consequence disclosure and nutrient-dense reformulation. (Dissemination detail is expanded in §18.)

8. Personal Objectives

Through this IRP I aim to strengthen a rigorous design practice at the intersection of behavioural design, speculative futures and ethical health innovation — in particular, translating ethical constraints (Food Freedom; ED-risk mitigation; de-moralisation) into actionable, evaluable design heuristics. A second objective is methodological: to practise research that is both critical and empirically grounded, using speculative artefacts to provoke critique while still generating analysable data. I want this project to position me for work in preventive-health and metabolic-health design.

9. Project Plan to End of Programme

A phased timeline runs May–August 2026: recruitment and protocol refinement (May–Jun), primary research and auto-ethnography synthesis (Jun–Jul), speculative prototyping (Jul), participatory critique and iteration (Jul–Aug), and final synthesis, RCA2026 exhibition build and report (Aug).



Part 2: IRP Realisation

10. Summary Statement

(Public-facing, third person)

Weijie Li's project asks why eating well feels like nothing. Unlike exercise or sleep, good nutrition returns almost no immediate, legible feedback — a structural gap that leaves well-intentioned people unable to connect what they eat to how their bodies actually fare. The project reframes this not as a willpower problem but as a **broken feedback infrastructure** resting on two opacities: we cannot easily see what processed food contains or does (*food opacity*), and we cannot sense our own metabolic state (*body opacity* — one can be metabolically unwell without being visibly overweight).

Through auto-ethnography, case studies and speculative design, the project identifies the decisive design move as **de-moralisation**: making the invisible visible through feedback that describes and navigates rather than scores and judges — preserving food joy and reducing the disordered-eating risk associated with punitive tracking.

The preferred future is a **de-moralised feedback infrastructure** that carries society from reactive Medicine 2.0 to proactive, personalised Medicine 3.0, modelled as a double helix in which user-side legibility and industry-side consequence-disclosure co-evolve. It is designed for a large, largely unaware metabolically-vulnerable public, with the diagnosed pre-diabetic cohort as its sharpest-need apex. Outputs include the Feedback Infrastructure Framework, the REFLECTIVE heuristic, a 2050 scenario set and speculative interfaces.

11. Key Research Findings



Workshop

Term 3 research and what it changed.

Term 3 research had four strands: an extended **auto-ethnography** of my own eating and metabolic self-observation; a set of **conversational and contextual interviews** conducted in late May–early June 2026 (a group picnic contextual-inquiry, and paired semi-structured interviews about eating, dieting and the body); **case-study analysis** of dietary-feedback products (MyFitnessPal, Noom, Oura, Levels/Zoe, Abbott Lingo) and food-systems policy; and continued literature review into metabolic health and food-systems science. The interviews were peer-recruited and partly informal — a sampling limitation I return to in §19 — but they triangulated the auto-ethnography strongly and reshaped the project’s framing.

Finding 1 — The feedback gap is two coupled opacities, not one. Auto-ethnography made vivid that the problem is not only *body opacity* (I cannot feel my glucose or visceral fat — the TOFI phenotype shows one can be “thin outside, fat inside” and metabolically unwell; Thomas and Bell, 2012) but also *food opacity* (ultra-processed products are engineered for cheapness, shelf-life, palatability and scale, with diffuse, delayed harms; Lancet UPF Series, 2025). The two compound: an illegible food meets an illegible body, so feedback is impossible *by construction*. This confirms and deepens the Proposal’s “phenomenological silence” while relocating part of the cause from the individual to the food environment.

Finding 2 — The harm in tracking is moral, not informational. Across my own logs and the case studies, the damage came not from data but from the **judgement**

layered on it: a glucose spike read as “I failed”, a dessert read as “today is ruined”. This both confirms the Proposal’s eating-disorder-risk literature (Roth et al., 2024; Wallace et al., 2025) and *contradicts* my earlier instinct to reject data outright. The resolution — **de-moralisation** — keeps legibility but strips judgement, reframing data as navigation (a sat-nav reroutes without scolding). This is the single most important conceptual shift of Term 3.

Finding 3 — The gap is systemic, so the response must be too. Mapping the problem revealed that individual feedback cannot win against a supply side optimised for opacity. This produced the **double-helix Feedback Infrastructure** (§14): a Demand Arm (personal visibility → actionable feedback → behaviour change → aggregate data) coupled to a Supply Arm (consequence disclosure → formulation change → cleaner ecosystem → policy support), with enablers (employers, government, culture). This is a genuinely *new perspective* relative to the Proposal, which treated feedback as an individual interaction problem.

Finding 4 — There is a defensible, high-need beachhead. Pre-diabetes is reversible, exploding in prevalence, and creates a legitimate, non-pathologising need to *see* food’s effect — making it the sharpest tip of a much larger metabolically-vulnerable, mostly unaware public (§13). This sharpens the Proposal’s two-cohort framing into an actionable audience.

Finding 5 — The phenomenological silence is confirmed, almost verbatim. Across interviews, participants independently restated the project’s founding intuition: that eating well produces “no feedback”; that after a healthy meal “the stomach is satisfied but the mouth and the brain are not”; and that the intense stimulation of ultra-processed food raises a “pleasure threshold” against which plain food then registers as nothing. This is strong triangulating evidence for the phenomenological-silence thesis — and it reframes the design target from *adding willpower* to *repairing a missing signal*.

Finding 6 — The harm the project must design against is already lived; and the demand for data is double-edged. The interviews surfaced a spectrum the project must protect against: from low-grade guilt and “remedial” compensatory eating after a “big” meal, to rigid food rules, to — at the most concerning end — accounts of multi-day fasting, compensatory over-exercise, and behaviour tied anxiously to the number on the scale, with food described as a “task” or a source of guilt rather than nourishment. Crucially, the *same* participants asked for **more** quantification — “just tell me how much to eat,” a tool that reveals what a food “really” contains, even continuous glucose data. This is the project’s central tension in miniature: the audience genuinely wants legibility, but unguarded legibility (scores, weights, calorie targets) is exactly the mechanism that produces the harm. The finding does not weaken the case for de-moralisation; it is the case for it — *deliver the legibility people ask for, without the judgement that hurts them*.

Finding 7 — Two design openings: commensality, and nutrient density. First, in the group picnic contextual-inquiry, people who restricted when eating alone ate

freely and without guilt in company — guilt was “absorbed into the social value” of the meal. This identifies **commensality and ritual** as a practical de-moralisation lever (echoing French and Japanese food cultures; Fischler, 1988; Rozin et al., 1999) and grounds the ambient/social scenario. Second, participants repeatedly named a “holy grail” wish — food that is enjoyable *and* nourishing without a metabolic penalty — and correctly diagnosed the modern problem as **high energy density, low nutrient density, and engineered palatability**. This points to a supply-side opportunity: reformulation toward **nutrient density and satiety-per-calorie**, paired with demand-side feedback that helps people match food to genuine physiological need rather than to a weight target. *(I deliberately frame this as nutrient density and satiety, not “low-calorie”: a weight-loss framing would re-import the diet-culture logic the project exists to resist.)*

Together these confirm the Proposal’s core (intention–action gap; ED-risk as a first-order constraint), evolve its stance (from rejecting data to de-moralising it), and add a systemic, audience, and supply-side dimension that directly shapes the outputs.

12. Project Development

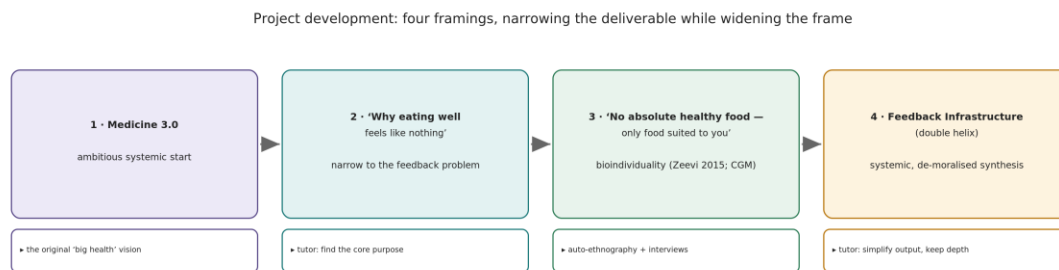


Figure 2 — Project development arc: four framings that progressively narrowed the deliverable while widening the frame, each pivot triggered by tutor critique and primary research.

The project’s shape changed substantially across the year, and the changes are themselves part of the research. The arc moved through four framings: **(1) Medicine 3.0** (an ambitious systemic start) → **(2) “Why eating well feels like nothing”** (narrowing to the feedback problem) → **(3) “No absolute healthy food — only food suited to you”** (bioindividuality; Zeevi et al., 2015) → **(4) The Feedback Infrastructure / double helix** (synthesis: a systemic, de-moralised reframe that re-integrates the Medicine 3.0 ambition at a meso scale). Each pivot narrowed the *deliverable* while widening the *frame*, in direct response to tutor critique to simplify the output without losing systemic depth.

13. Audience

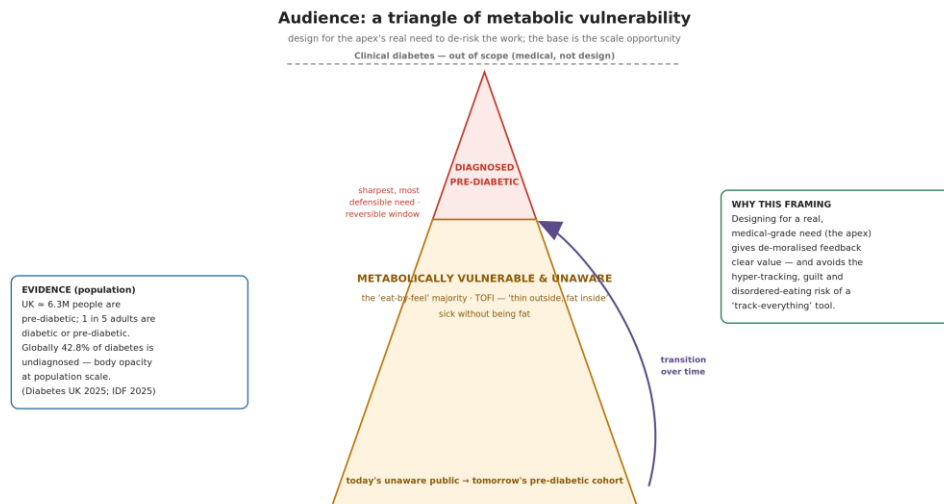


Figure 3 — The audience as a triangle of metabolic vulnerability: a narrow apex of diagnosed pre-diabetic people (sharpest, reversible need) above a broad base of metabolically-vulnerable, unaware people (the 'eat-by-feel' majority; TOFI). Clinical diabetes sits above the line, out of scope.

The audience is best drawn as a **triangle of metabolic vulnerability**. At the narrow **apex** sit people who have been *diagnosed pre-diabetic*: their need is the sharpest and most defensible — pre-diabetes is reversible, and they have a legitimate, non-pathologising reason to see what food does to them. Beneath them, forming the broad **base**, is the much larger population of **metabolically-vulnerable but unaware** people — busy individuals who “eat by feel”, and the TOFI (“thin outside, fat inside”) group who look fine but carry hidden risk (Thomas and Bell, 2012). The base is the apex’s *upstream pipeline*: today’s unaware public is tomorrow’s pre-diabetic cohort unless the trajectory is interrupted. Crucially, the project **stops short of diagnosed diabetes**, which is a clinical/medical domain; staying on the *pre-clinical* side keeps the work preventive and design-led rather than medical.

This framing is both an opportunity and a safeguard. Designing for the apex’s hard, real need *de-risks* the design ethically: where there is a genuine reason to make food’s effects legible, de-moralised feedback has clear value — and by serving that need carefully we avoid manufacturing the hyper-tracking, anxiety and disordered-eating risk that a “track everything, for everyone” tool would create. The base, meanwhile, is the scale opportunity and the public-health prize.

The evidence is stark. In the UK, an estimated **6.3 million people** have pre-diabetes (non-diabetic hyperglycaemia) and **one in five adults** now lives with diabetes or pre-diabetes (Diabetes UK, 2025). Globally, **12.0% of adults (≈635 million)** had impaired glucose tolerance in 2024, and **42.8% of people with diabetes are**

undiagnosed — a direct measure of body opacity at population scale (IDF, 2025). Most of the base does not know which triangle they are in.

14. Complexity Map — The Double-Helix Feedback Infrastructure

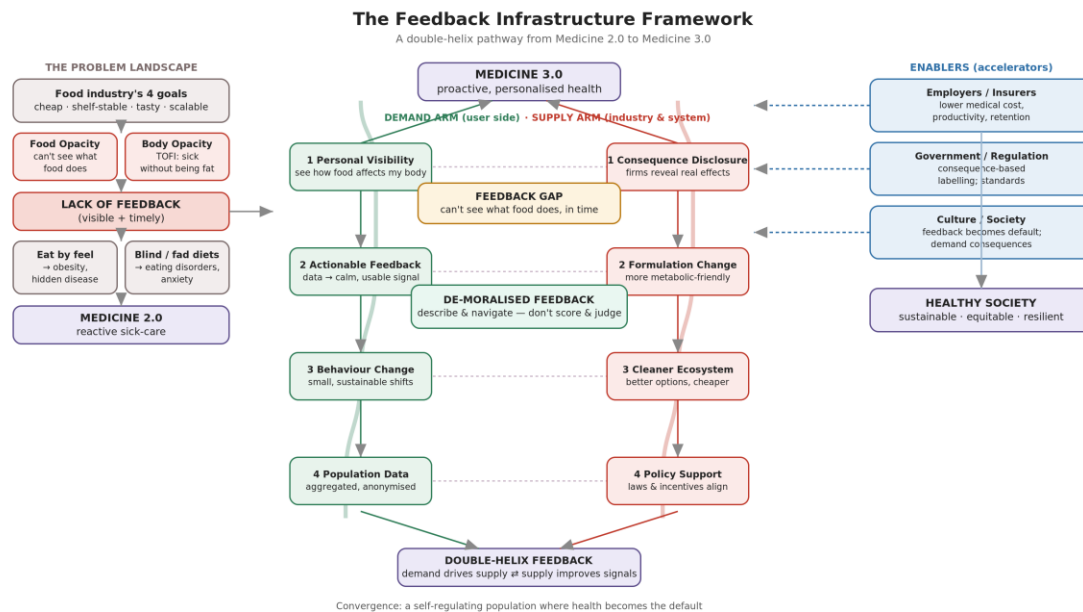


Figure 4 — The Feedback Infrastructure Framework: a double-helix pathway from Medicine 2.0 to Medicine 3.0. Left: the present problem landscape (food opacity + body opacity → a lack of visible, timely feedback → Medicine 2.0 sick-care). Centre: the Demand Arm (green) and Supply Arm (red) coiled around the central Feedback Gap, with de-moralised feedback as the core design move, converging upward to Medicine 3.0. Right: the enablers (employers, government, culture) that accelerate the helix toward a healthy society.

What this map shows, and what it leaves out. The map renders the *feedback infrastructure* around dietary behaviour as a single system, deliberately spanning three zones that are usually analysed separately: the **problem landscape** (left), the **double-helix mechanism** (centre), and the **enablers** (right). It is intentionally *not* a clinical-treatment map, a technical architecture, or a product spec — those are left out so the structural relationships stay legible.

The central tension is the **Feedback Gap**: *people cannot see what food does to their bodies in time*. It is produced on the left by the Standard American Diet environment and the food industry's four goals (cheap, shelf-stable, tasty, scalable), which generate *food opacity*; by nutrition being multi-dimensional, individual, and effectively a privilege of wealth and time; and by *body opacity* (difficulty sensing internal response). The result is a lack of visible, timely feedback, which splits into two failure modes — *eating by feel* → obesity and hidden disease (“sick without

being fat”), and *blind/fad diets* → eating disorders and anxiety — both sustained under reactive **Medicine 2.0**.

The double helix is the proposed repair. A **Demand Arm** (Personal Visibility → Actionable, non-anxious Feedback → small, sustainable Behaviour Change → aggregated, anonymised Population Data) winds around a **Supply Arm** (Consequence Disclosure → Formulation Change → a Cleaner Food Ecosystem → System & Policy Support). The two are coupled — **demand for legible consequence drives better supply; better supply improves the signals available to demand** — so the loops reinforce rather than compete, converging toward a self-regulating population in which health becomes the default: **Medicine 3.0**. On the right, three **enablers** accelerate the helix: *employers/insurers* (lower medical cost, productivity, retention), *government/regulation* (consequence-based labelling; turning voluntary action into standards), and *culture* (feedback becomes the new normal; people demand consequences, not just data).

15. Scenarios

Time horizon: 2050 (with a 2035 waypoint). I set the primary horizon at **2050** — roughly one generation out. This is justified on three grounds: it is the maturation horizon for Medicine 3.0 infrastructure (Attia, 2023); it is the timescale over which today’s unaware metabolically-vulnerable *base* (§13) develops, or avoids, clinical disease — so the stakes are concrete; and it is far enough to denaturalise present diet-culture assumptions while remaining actionable by backcasting. A nearer **2035** waypoint marks the consumer-CGM transition already underway.

Following Dator’s alternative futures and the futures cone (Dator, 2009; Inayatullah, 2008), I use four scenarios as **critique instruments, not predictions** — devices to let stakeholders “try on” feedback logics and articulate preference, fear and ethical limits.

- **S0 — Diet-Culture Lock-in (*Discipline / Collapse; the cautionary baseline*).** Engineering-led, optimisation-first dashboards become the default vocabulary of metabolic feedback. Measurement becomes continuous judgement; scores, streaks and “good/bad” foods harden; eating-disorder and anxiety rates climb. This is the future the project exists to prevent.
- **S1 — Food as Code (*Growth; probable but risky*).** Continuous biometric optimisation delivers dazzling personal control, but “wellness” interfaces re-moralise data and reify the body into a dashboard. Desirable to early adopters; corrosive at population scale.
- **S2 — The Disappearing Interface (*ambient/social; desirable but exposed*).** The environment — tableware, lighting, sound, social pacing — delivers post-meal feedback indirectly, echoing French/Japanese commensal cultures (Fischler, 1988; Rozin et al., 1999). Low anxiety, but vulnerable to ambient surveillance and manipulation without consent.
- **S3 — The De-Moralised Navigator (*Transform; preferred*).** Technology makes food and body legible **without judgement**, scaffolds interoceptive

literacy, and then recedes as skill grows — “the tool that disappears”. The risk is commercial incentive to prevent off-boarding and maintain dependence. This is the future developed in §16, illustrated below.

Illustrative vignette — Theo’s day, London, 2050 (Scenario S3). *To make the preferred scenario concrete, a single day with a fictional user, Theo (28, a design strategist), experienced through three de-moralised interface layers:*

Morning — the Strategic layer. While washing, Theo glances at a **Healthspan Curve** on the mirror — no weight, no calorie balance. It shows his cardiovascular and neurodegenerative *resilience*: weeks of swapping in monounsaturated fats has improved his apoB clearance, pushing his cardiovascular-risk curve two years to the right. The feedback is *compounding* and *long-horizon*, and it produces calm, not anxiety.

Midday — the Tactical layer. In a shop, a lightweight AR lens overlays a **Food Architecture Radar** on each item. A “healthy” juice shows a sharp insulin-spike polygon; an avocado-and-protein salad shows a flat, durable one. There is no “good food / bad food” verdict — Theo simply picks the block that fits the next four hours of focused work, “like a craftsman choosing a part”.

Night — the Neutral Readable layer. At a flatmate’s birthday he eats a large slice of chocolate cake. The old loop — “today is ruined; I have no willpower” — does not start. Instead the system quietly describes the physical reality without scoring it and, like a sat-nav after a wrong turn, offers a few equally-valid routes: move a little more tomorrow; take a short walk now; or simply do nothing, because one meal is not a verdict. Theo carries on, without moral panic, because the strategic layer shows his long-term resilience is intact.

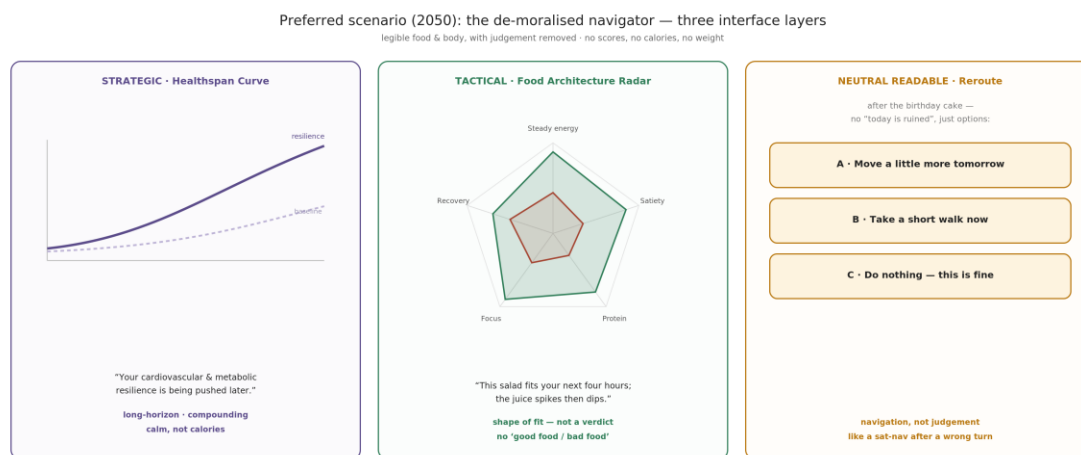


Figure 5 — The preferred 2050 scenario, “the de-moralised navigator”, across three interface layers: a Strategic Healthspan Curve (resilience over time), a Tactical Food Architecture Radar (the shape of how a food fits the next few hours — not a good/bad verdict), and a Neutral Readable reroute (options after an indulgent meal, framed as

navigation rather than judgement). The interface carries no scores, calorie counts or weights — de-moralisation made concrete.

16. Preferable Futures

The preferred future is **S3 — a de-moralised feedback infrastructure**: one in which food and body are made *legible in time*, but legibility never tips into judgement. It is the maturation of the double helix (§14) — a world where the Demand Arm (people can finally see consequence) and the Supply Arm (industry must disclose it) have co-evolved far enough that *health has become the default*, the definition of Medicine 3.0. Theo's day (§15) is its lived texture: data that orients rather than disciplines; a body made readable without being reified; choices — including “indulgent” ones — made from information and self-trust rather than guilt.

For whom is this preferable, and why. For the metabolically-vulnerable *base* and the pre-diabetic *apex* (§13), it converts an invisible, slow-moving risk into something seeable and reversible *without* the anxiety that makes current tools unsafe. For people vulnerable to disordered eating, de-moralisation removes the specific mechanism — moral judgement on data — that the literature links to harm (Roth et al., 2024; Wallace et al., 2025). For industry, consequence disclosure plus aligned incentives (employers, regulation) make metabolic-friendly formulation *competitive* rather than charitable. For society, a self-regulating population lowers the chronic-disease burden that strains health systems.

Limitations and further development. The future is not automatically benign. The same infrastructure that makes consequence legible can enable **surveillance and behavioural manipulation**, and raises hard **equity** questions — who owns the data, and does precision prevention widen or narrow the health gap that already makes good nutrition a privilege? “De-moralisation” is also fragile: markets and culture re-moralise easily (“clean eating 2.0”), so the principle needs active defence. The “tool that disappears” runs against commercial incentives to retain users — which is why **designed obsolescence** must be an explicit principle: like training wheels, the tool should actively work to make itself unnecessary, building bodily intuition and then receding, rather than maximising engagement. A bounded arc (for example, around 90 days) with a measurable “graduation” is one way to design the exit in. And the model still needs **primary validation**: the auto-ethnography and case studies motivate it, but participatory testing of de-moralised feedback with the target audience is the necessary next step.

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Appendices

The detailed appendix materials carried over from the IRP Proposal submission — Appendix A (Annotated Literature and Practice Review), Appendix B (Signals / Trend Wall), Appendix C (Ethics Checklist), Appendix D-E (Research Protocols, method cards M1-M7), Appendix F (2035 Scenario Narratives), Appendix G (The After Glow critical-design specifications) and Appendix H (supporting Data Visualisations) — are appended on the following pages.

Appendix pages (carried over from the IRP Proposal submission)

The following pages reproduce the original appendices: annotated literature & practice review, signals, ethics checklist, method cards M1-M7, 2035 scenario narratives, The After Glow specifications, and supporting data visualisations.

A Statement of Authorship, Acknowledgements, and AI Disclosure

This proposal was written independently as part of the IRP Proposal module (Global Innovation Design, Royal College of Art). The research direction emerged from an ongoing inquiry into preventive health, dietary behaviour, and the design of feedback systems.

I am grateful to tutors and peers for feedback during the April 2026 project discussion and critique, which sharpened the ethical focus on eating-disorder risk and food freedom as constraints on any proposed dietary feedback design.

Use of AI disclosure

This submission was produced by me in my own words and using my own imagery, except for quotations and images from published and unpublished sources, which are clearly indicated and acknowledged. AI assistance was used for (i) systematic literature identification to locate relevant behavioural science papers, and (ii) technical document formatting and data visualisation rendering. All critical analysis, theoretical framing, and speculative design directions are my original work.

B Appendix A: Annotated Literature and Practice Review

Selected annotations (What / So what / Now what) are summarised below; the original long-form draft exists in `main.tex`.

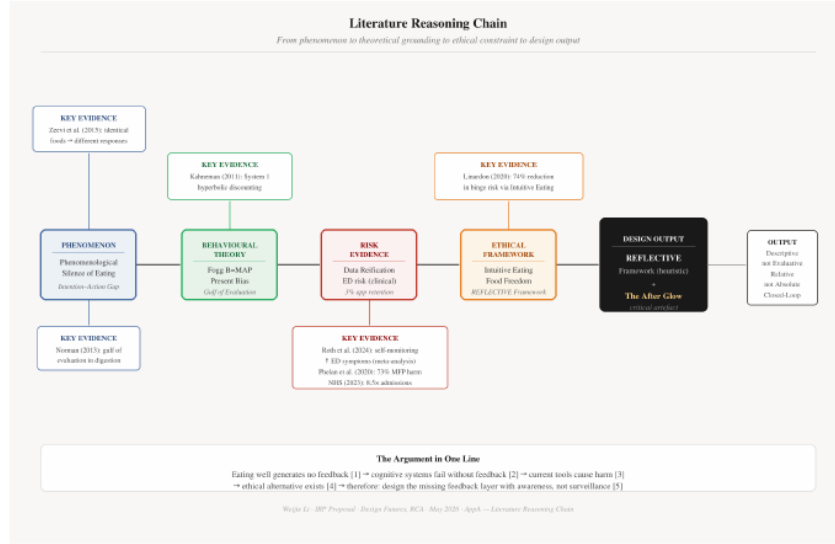


Figure A: Literature logic chain used to connect the feedback-gap problem framing to harm pathways and design criteria.

- **Kristeller and Wolever (2011) — MB-EAT.** *So what:* an evidence-based alternative to calorie counting grounded in interoceptive awareness. *Now what:* supports an awareness-first feedback orientation.
- **MyFitnessPal (practice).** *So what:* high-friction, evaluative self-monitoring with documented risk pathways. *Now what:* motivates rejecting manual logging/scoring as a default interaction paradigm.
- **Oura (practice).** *So what:* makes invisible sleep signals legible, but via evaluative scoring. *Now what:* borrow “make the invisible felt” while avoiding judgement.
- **Zeevi et al. (2015).** *So what:* bioindividuality undermines population-average dietary rules. *Now what:* grounds “relative to baseline” feedback.
- **Wallace et al. (2025).** *So what:* articulates reification of data as a harm mechanism. *Now what:* informs REFLECTIVE’s “descriptive not evaluative” constraint.

C Appendix B: Signals / Trend Wall

Three signal cards (abridged) are retained below. If a signal’s supporting image was deleted as out-of-date, it will appear as a missing-file placeholder where referenced.

Signal 1 — The OTC CGM inflection point

The June 2024 FDA over-the-counter clearance of Abbott Lingo signals a shift of metabolic tracking from clinical infrastructure to consumer wellness. Design risk: engineering-led evaluative dashboards may become the default interaction vocabulary.

Signal 2 — Intuitive Eating and the anti-diet movement

Intuitive Eating is mainstreaming interoceptive literacy as a protective framework against diet culture. Design implication: feedback should direct attention back to bodily authority, not outsource it to scores.

Signal 3 — Social ritual as non-technological feedback

Cross-cultural practices show non-punitive feedback without surveillance (e.g., gratitude rituals, communal pacing, environmental cues). Design implication: technology can scaffold rituals rather than replace them with metrics.

D Appendix C: Ethics Checklist

The submitted ethics checklist is included below for completeness.



Outlook

Thanks for filling in this form: RCA: Ethics Checklist

发件人 Forms response receipts <forms-receipts-noreply@google.com>

日期 周二 2026/4/14 17:46

收件人 10066513@network.rca.ac.uk <10066513@network.rca.ac.uk>

Thanks for filling in this form: [RCA: Ethics Checklist](#)

Here's what was received.

RCA: Ethics Checklist

This Ethics Checklist helps taught students at the RCA to identify the ethical risks involved in their proposed projects, and the steps needed to address those risks. This includes students studying on MA, MDes, MEd, MFA, MRes and Graduate Diploma programmes, as well as on AcrossRCA (College-wide unit).

The form should take no more than five minutes to complete.

Your email address (**10066513@network.rca.ac.uk**) was recorded when you submitted this form.

Name

Weijie Li

Does your project require ethical approval from an external agency, such as the NHS, a partner institution or an employer?

*

☐ Yes

☒ No

If yes, please specify who

Does your project involve human participants (e.g. it involves interviews, surveys, workshops or user-testing?)

*

☒ Yes

☐ No

Human participants

If your research project involves human participants you will need to develop an information sheet that describes the nature of your project and a participant consent form that provides a record of what participants are agreeing to. The template is available on [this page](#). If you are working with a third party to secure access to participants you may also have to develop a gatekeeper consent form.

Low risk questions

Does your project:

*

Yes

No

involve potential conflicts of interest between you and your participants

☒

☐

create outputs that identify individuals and/or organisations?

☒

☐

Medium risk questions

Does your project: *

	Yes	No
involve discussion of topics that participants may find distressing?	☐	☒
involve the offer of financial incentives to participants?	☐	☒
involve the collection of data without the informed consent of participants?	☐	☒
risk disclosing the physical and/or mental health of participants?	☐	☒

High-risk questions

Does your project: *

	Yes	No
involve participants under 18 years of age?	☐	☒
involve participants who lack the capacity to consent?	☐	☒
involve medical research with humans?	☐	☒
involve the administration of drugs, placebos or other substances (e.g. food, vitamins) to humans?	☐	☒
risk causing physical and/or psychological harm to participants?	☐	☒

Did you answer yes to any **medium** or **high-risk** questions relating to human participants? *

- ☐ Yes
- ☒ No

Other risks

Medium-risk questions

Does your project: *

	Yes	No
take place outside of the UK?	☐	☒
involve the collection, use or storage of non-living biological samples not already held in a university/museum collection?	☐	☒
involve the collection, use or storage of human tissue/DNA?	☐	☒
Involve materials and/or elements that are under threat or endangered?	☐	☒
Involve materials or substances that are toxic to the environment?	☐	☒
Risk causing other forms of damage to the environment (beyond normal daily activity)?	☐	☒

High-risk questions

Does your project: *

	Yes	No
involve animal participants?	☐	☒
risk disturbing or damaging culturally, spiritually or historically significant artefacts/places/remains?	☐	☒
relate to the defence industry?	☐	☒
involve potentially illegal activities?	☐	☒

Did you answer yes to any **medium** or **high-risk** questions relating to other risks? *

- ☐ Yes
- ☒ No

Lower-risk projects

Your project has been identified as a lower-risk project. You can continue with your project in consultation with your tutor without seeking further ethics approval.

You can read more about identifying and addressing ethical risks on [this page](#).

☒ End checklist

[Create your own Google Form](#)

Does this form look suspicious? [Report](#)

E Appendix D: Research Protocols (Phases 0–5)

This appendix provides the method overview cards used as participant-facing prompts. Each card summarises the intent, structure, and outputs of one method.

M1 — Probe Kit

The probe kit primes participants to notice post-meal sensations and the “silence” of dietary feedback before any speculative artefact is introduced.

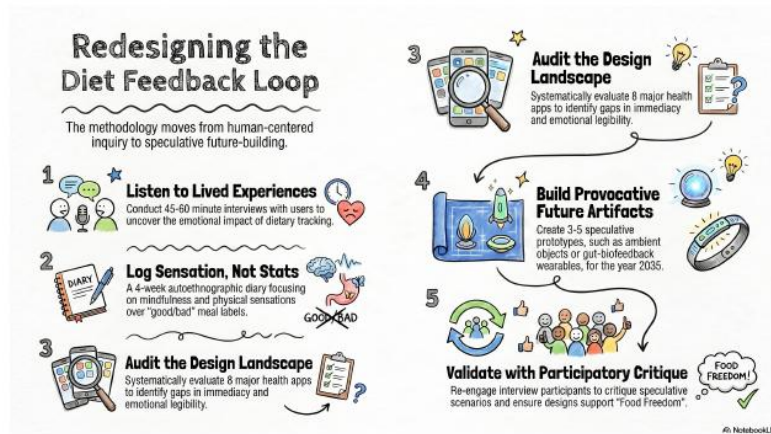


Figure B: M1 method card: Probe Kit.

M2 — Photo Elicitation

Participants capture the aftermath of meals to make “what happens after eating” discussable without focusing on food itself.

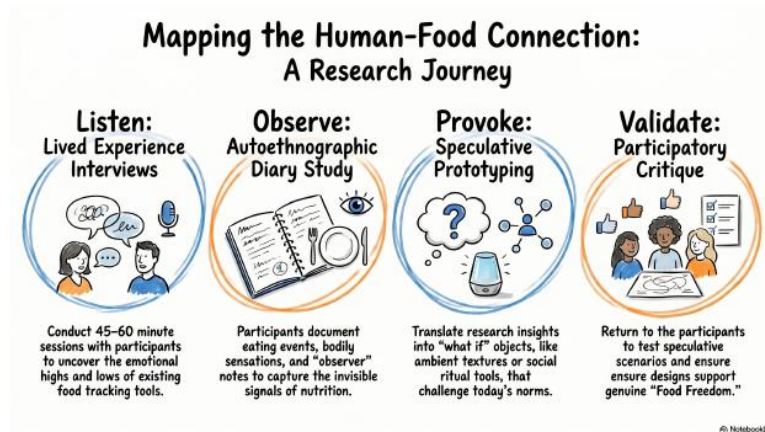


Figure C: M2 method card: Photo Elicitation.

M3 — Contextual Inquiry

In-situ observation identifies friction points and existing informal feedback practices in real eating contexts.

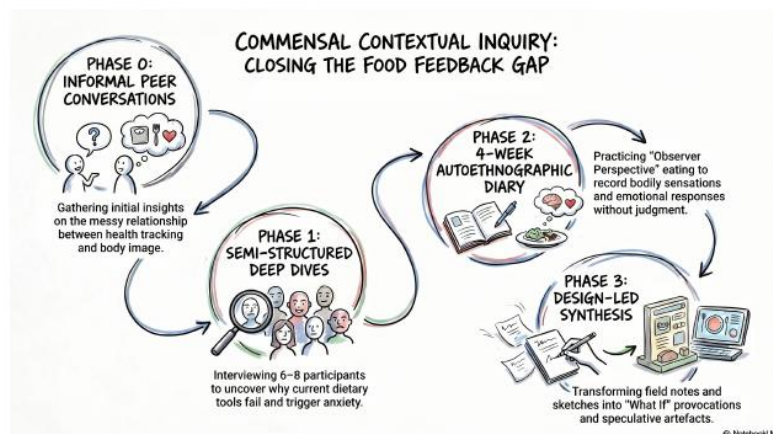


Figure D: M3 method card: Contextual Inquiry.

M4 — Ecological Momentary Assessment (EMA)

Short post-meal prompts sample experience across days to validate the feedback gap beyond single interviews.

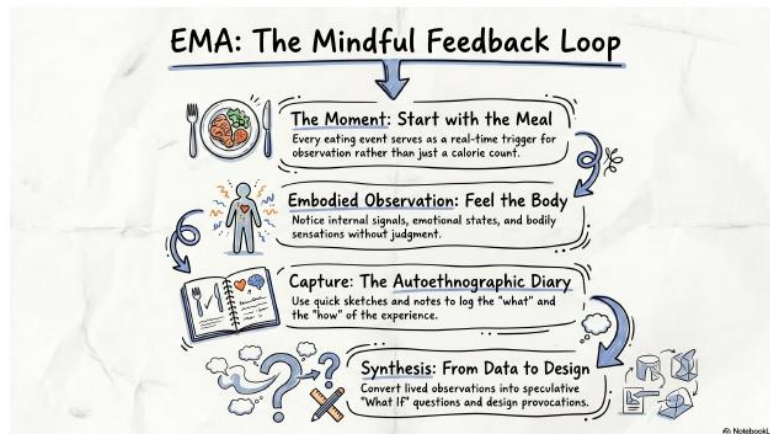


Figure E: M4 method card: Ecological Momentary Assessment (EMA).

M5 — Mindful Diary

A diary protocol captures somatic vocabulary and converts lived moments into design provocations (non-numeric, non-evaluative).

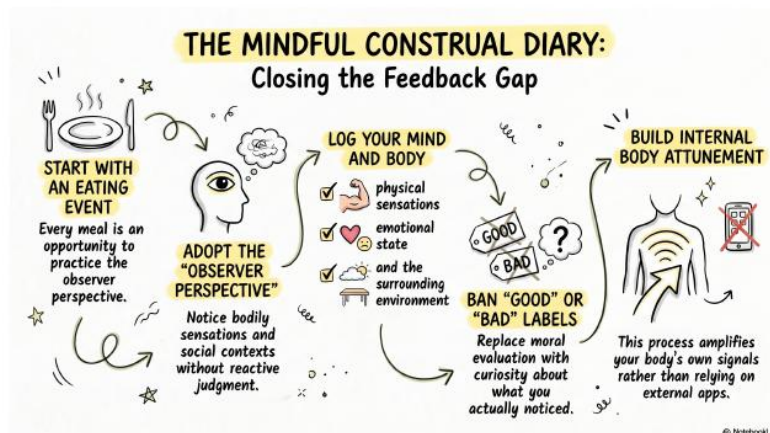


Figure F: M5 method card: Mindful Diary.

M6 — Body Mapping

Body mapping establishes a shared, non-verbal vocabulary for internal sensations before critique activities.

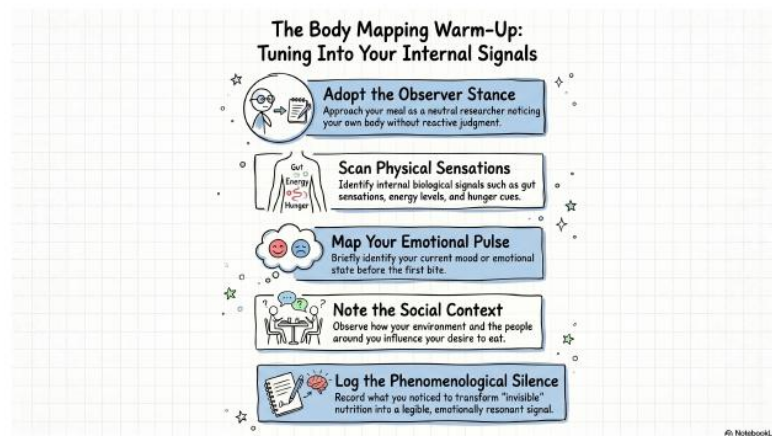


Figure G: M6 method card: Body Mapping.

M7 — Anti-Heroes

Adversarial personas surface ethical failure modes and protect against “compassion-washing” of punitive tracking, drawing on work on folk theories of tracking and algorithmic systems. (Makhortykh et al., 2022)

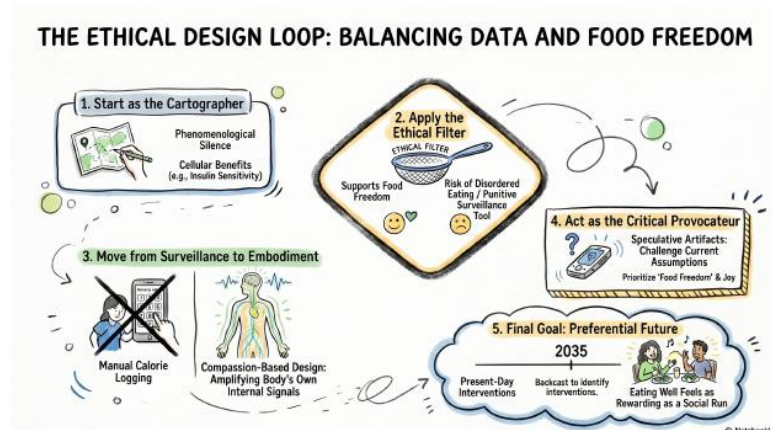


Figure H: M7 method card: Anti-Heroes.

F Appendix E: 2035 Scenario Narratives

This appendix summarises the three scenarios used as critique stimuli. They are not forecasts; they are structured provocations used to surface values, fears, and ethical boundary

conditions.

Scenario 1 — Food as Code (high-tech biometric)

A near-future in which dietary feedback is dominated by continuous metabolic optimisation and evaluative scoring. *Primary risk*: total reification of data and diet-culture re-inscription through “wellness” interfaces.

Scenario 2 — The Disappearing Interface (ambient social)

A near-future in which the environment (tableware, lighting, sound) delivers post-meal feedback indirectly via ritual and ambience. *Primary risk*: ambient surveillance and manipulation without consent.

Scenario 3 — The Tool That Disappears (intuitive technology; preferred)

A near-future in which technology is a temporary scaffold for interoceptive literacy: it prompts attention, then reduces itself as skill develops. *Primary risk*: commercial incentives to prevent off-boarding and maintain dependence.

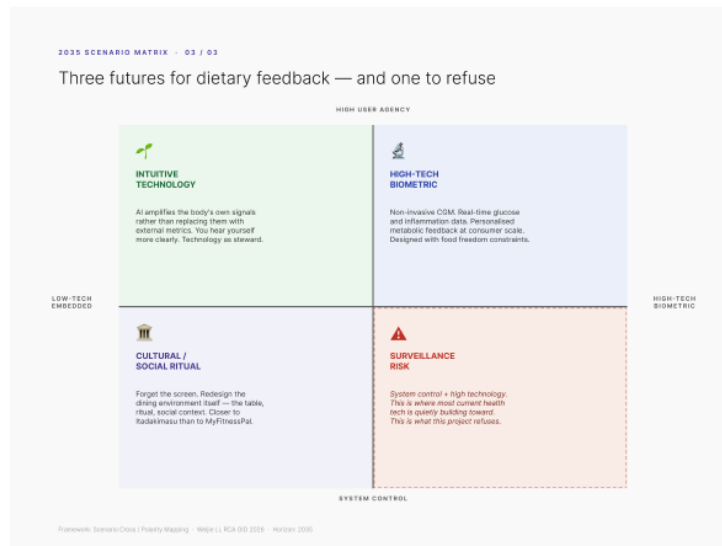


Figure I: Scenario matrix used to structure the 2035 scenario set.

G Appendix F: The After Glow — Critical Design Specifications

This appendix documents the artefact rationale and design intent. Visuals below support the distinction between descriptive, bounded rituals and evaluative, open-ended surveillance.

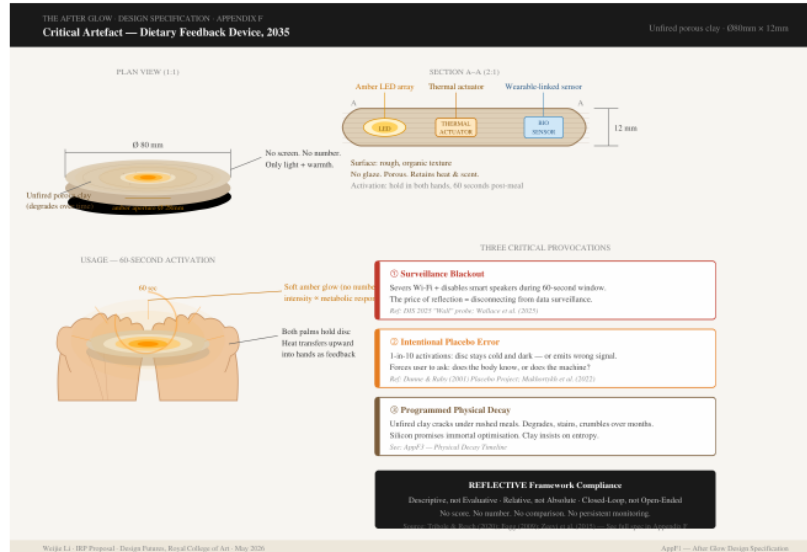


Figure J: After Glow: early design sketch (form factor and interaction ritual).

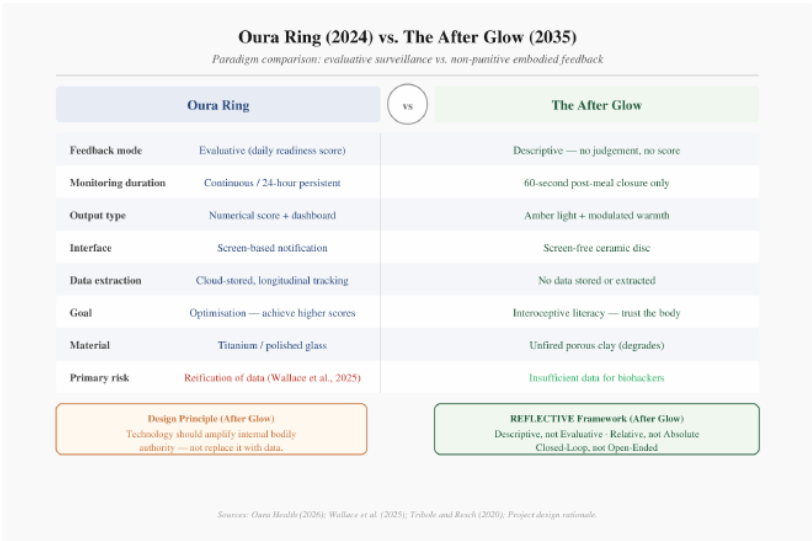


Figure K: Oura-style evaluative dashboards versus After Glow’s descriptive, bounded post-meal ritual.

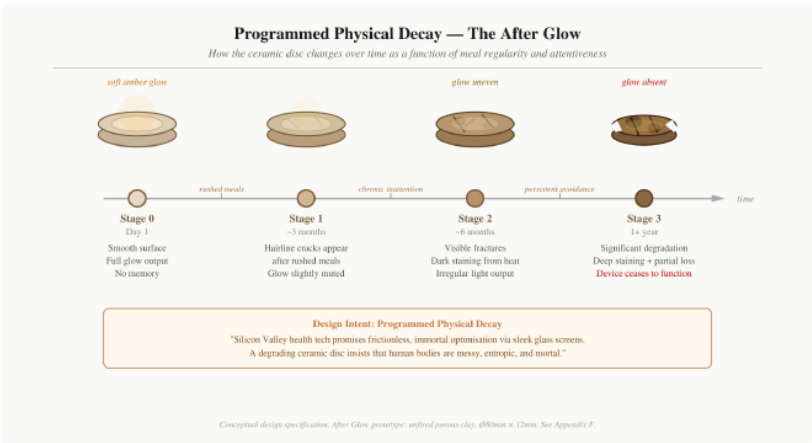


Figure L: Physical decay timeline: material temporality and maintenance logic as part of the critique.

H Appendix G: Data Visualisations

This appendix collects supporting figures referenced in the main argument. Each figure is placed immediately after a brief statement of what it evidences.

G1 — Health app retention

This figure supports the claim that prevailing self-tracking paradigms struggle to sustain engagement beyond the first month. (Business of Apps, 2024)

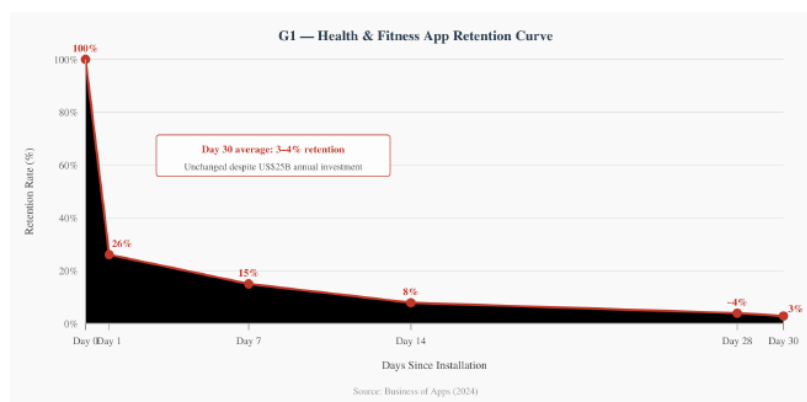


Figure M: Health/fitness app retention curve (Day 0–Day 30).

G2 — OTC CGM market growth

This figure situates the work in the near-term shift of metabolic sensing from clinical settings into consumer products. (Precedence Research, 2025)

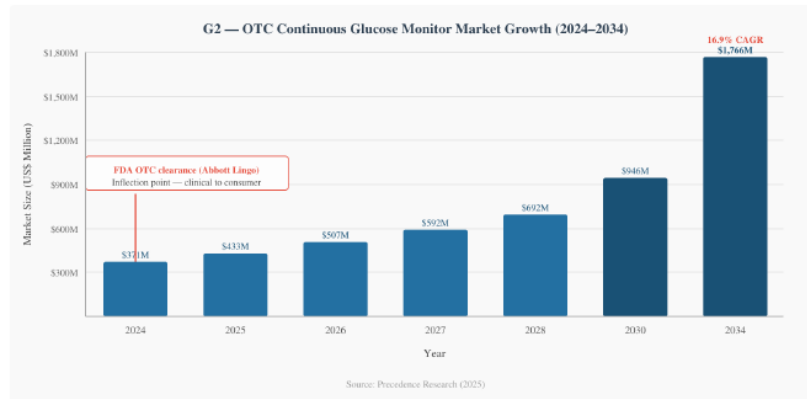


Figure N: OTC continuous glucose monitoring (CGM) market growth signal.

G3 — Eating-disorder admissions

This figure provides contextual urgency for harm-prevention constraints when designing dietary feedback. (NHS England, 2023)

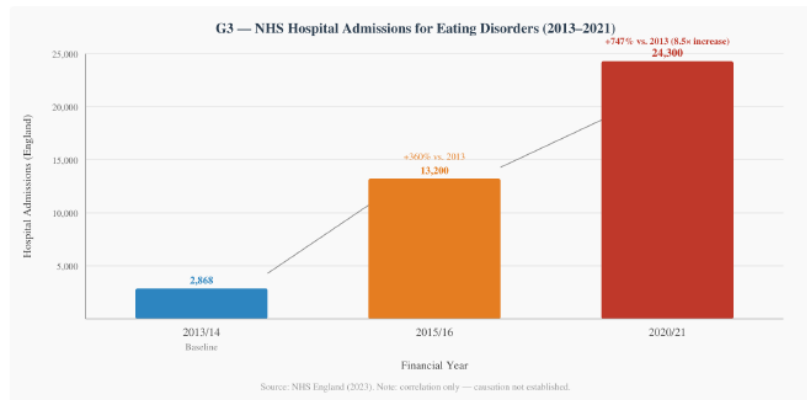


Figure O: NHS eating-disorder hospital admissions trend.

G4 — Intuitive eating as protective association

This figure supports an “awareness-first” direction by visualising the protective association reported in the literature. (Linardon, 2020)

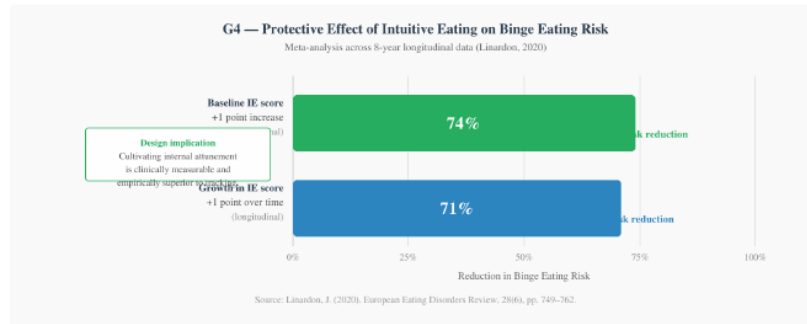


Figure P: Intuitive eating as a protective association signal (e.g., reduced binge-eating risk).

G5 — Translating principles into constraints

This figure maps the Intuitive Eating principles into concrete design constraints used to evaluate feedback concepts. (Tribble and Resch, 2020)

🔑 Key Principles & Design Implications

Intuitive Eating Principle	Design Constraint for This Project
Reject the Diet Mentality	No weight-loss framing in feedback
Honour Your Hunger	System must not override internal hunger signals
Make Peace with Food	No "good food / bad food" binary in any feedback
Challenge the Food Police	Eliminate guilt-inducing notifications or scoring
Discover the Satisfaction Factor	Feedback should reinforce pleasure, not diminish it
Feel Your Fullness	Design should amplify, not replace, body attunement
Cope with Your Emotions without Using Food	System must not weaponise emotional states
Respect Your Body	No body-shaming metrics or comparisons
Movement — Feel the Difference	Feedback should feel good, not punitive
Honour Your Health with Gentle Nutrition	Nutrition guidance, not nutrition surveillance

Overall Assessment	Detail
Strength	Empirically supported; defines what "harm-free" design looks like
Limitation	Primarily a clinical/therapeutic framework — needs translation to product design

Figure Q: Intuitive eating principles translated into design constraints (working visual draft).

Peer Reviewer

Your full name: Wenjie Li (Oliver)

Assessing:

Project name: Designing for Digital Legacy especially in China's platform

Student name: Jinjun Shen

Ecosystem

Critically reflect and provide comments on the questions below.

What verbal feedback have you provided to your peer at critiques on April 14? Provide answers to the following questions based on your critical reflection. Once you complete it, take a photo and submit it on Moodle by 6pm on April 14. Hand a copy of this sheet to the student you are assessing on the same day.

a) Have they met / not met the learning outcomes of the module? Why?

The learning outcomes are:

- Identify a design futures topic that addresses an environmental and/or societal challenge, including situating the topic and its significance in relation to existing work and developing an appropriate research question;
- Propose a design futures project, including the application of relevant approaches and methods.

Please provide an answer (minimum 100 words)

- The students met the module's learning outcomes by identifying a critical design future topic: managing digital legacy in Chinese platform ecosystem.
- Problem Formulation: Highlighted the power imbalance where tech giants control data, leaving grieving families without access to non-financial assets.
- Appropriate methodology: proposed a well-rounded design futures approach using literature reviews, systematic mapping and scenario building.

b) What was the quality of their submission, and why?

Please provide an answer (minimum 100 words): The quality of submission was excellent.

- Effective visual: used "hero images" such as digital will and unlocked digital assistant, to successfully translate abstract future concepts into tangible design intervention.
- Systematic Analysis: Effectively categorized existing digital service (platform governance, AI afterlife) to ground their project in current realities.

c) What are the areas for improvement of their IRP Proposal, and how to improve?

Please provide an answer (minimum 100 words)

- clear structure: logically mapped - while the proposal is strong, there are specific areas for improvement regarding stakeholder research and ethical boundaries. Recommendations for improvement:
- Broaden stakeholders: Accessing digital assets currently requires legal clearance. The students should interview policy makers, legal experts, or platform representatives, rather than solely focusing on youth users.
 - Define privacy boundaries: The system must incorporate robust mechanisms for users to pre-determine what is inherited versus permanently deleted, ensuring clarity after death.
 - Address AI Ethics: The proposal should critically analyze the ethical implication of tech companies retaining data to create "digital versions".